

4	(a)	(i)	use of tangent at time $t = 0$ acceleration = $42 \pm 4 \text{ cm s}^{-2}$	B1 A1	[2]
		(ii)	use of area of loop distance = $0.031 \pm 0.001 \text{ m}$ allow 1 mark if $0.031 \pm 0.002 \text{ m}$)	B1 B2	[3]
	(b)	(i)	$F = ma$ $= 0.93 \times 0.42$ {allow e.c.f. from (a)(i)} $= 0.39 \text{ N}$	C1 A1	[2]
		(ii)	force reduces to zero in first 0.3 s then increases again in next 0.3 s in the opposite direction	B1 M1 A1	[3]

Page 2	Mark Scheme	Syllabus	Paper
	GCE A/AS LEVEL – November 2005	9702	2

5 (c) similarity as same wavelength/frequency/period, constant